

- 4 (a) Explain what is meant by an *ideal gas*.

An ideal gas is a theoretical gas that obeys the equation of state $pV = nRT$ at all pressures p , volumes V and thermodynamic temperatures T for a fixed mass of gas. R is the molar gas constant and n is the amount of gas in moles. [1]

.....[1]

- (b) Two vessels X and Y of volumes $10.0 \times 10^{-4} \text{ m}^3$ and $3.0 \times 10^{-4} \text{ m}^3$ are connected by a tube of negligible volume and kept at temperatures 200 K and 100 K respectively. Assume both vessels contain the same monatomic ideal gas.

Calculate the ratio of $\frac{\text{number of moles of gas in X}}{\text{number of moles of gas in Y}}$.

At steady state,

Pressure in X = Pressure in Y

Since $PV = nRT$ [1m for ideal gas equation and showing P are the same]

$$\frac{PV_X}{PV_Y} = \frac{n_X RT_X}{n_Y RT_Y}$$

$$\frac{n_X}{n_Y} = \frac{V_X T_Y}{V_Y T_X}$$

$$\frac{n_X}{n_Y} = \frac{10.0 \times 10^{-4}}{3.0 \times 10^{-4}} \times \frac{100}{200} = \frac{5}{3} = 1.67 \quad [1]$$

ratio = [2]

- (c) An ideal gas in a container with a movable piston is heated. At the same time, the volume is increased such that the temperature of the gas always remains constant. By considering the First Law of Thermodynamics, explain why the temperature of the gas remains constant even though it is heated.

The First Law of Thermodynamics states that the increase in internal energy ΔU is the sum of heat provided to the system Q and the work done on the system W . [1]

If heat supplied Q is equal work done by gas ($W = -Q$), ΔU is zero. Since temperature of gas is proportional to internal energy, $T \propto U$, the temperature remains constant. [1]

.....[2]

Also accept $\Delta T \propto \Delta U$

- (d) Fig. 4.1 below shows how the pressure p of the gas varies with its volume V in part (c). The volumes of the gas at initial and final states are V_A and V_B respectively. The pressures of the gas at initial and final are p_A and p_B respectively.

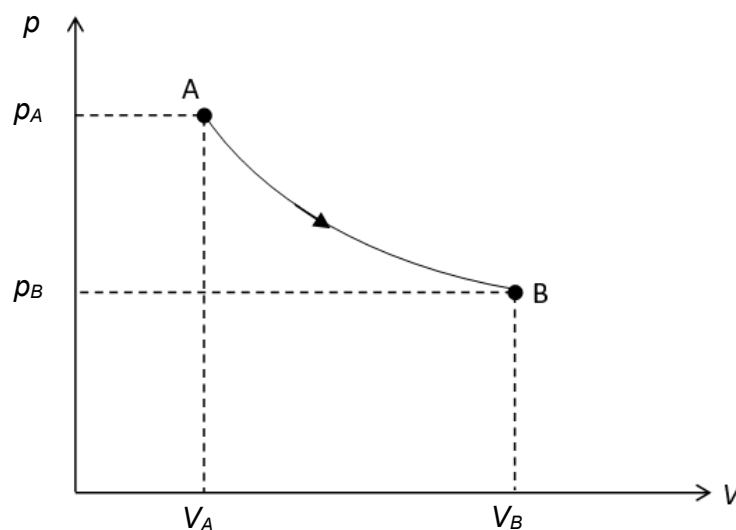


Fig. 4.1

The container in (c) is now insulated. The volume of the gas is increased to V_B again.

- (i) Use the First law of Thermodynamics to explain whether the final pressure is higher or lower than p_B .

The piston is insulated, thus it is an adiabatic expansion, $Q = 0$. [1]

Since it is an expansion, W is negative. By First Law of Thermodynamics,

$\Delta U = Q + W$, W is negative, $Q = 0$, thus ΔU is negative. Hence temperature falls. [1]

By ideal gas equation $PV = nRT$ [1], for the same final volume but at a lower temperature, the pressure must be lower than p_B . [1]

[3]

- (ii) Sketch, on Fig. 4.1, a graph to show the variation with volume of pressure of the gas as its volume increases in the insulated container. [1]

Pressure drops below p_B , with correct shape and direction of arrow clearly indicated.

[Total: 9]

5 (a) The variation with time t of the potential difference V , across a resistor is shown in Fig. 5.1